



Office of the HEAT Sub Project

PIN-15012

School of Engineering and Physical Sciences
North South University,
Bashundhara, Dhaka-1229



Technology Transfer Management System (TTMS)

Website & Intellectual Property Management Portal Development Specification

Document Type	Website & IP Management System Requirements Specification
Client	Technology Transfer Office (TTO), North South University
Version	1.0

01 PROJECT OVERVIEW

The Technology Transfer Office (TTO) requires the development of a centralized Technology Transfer Management System (TTMS) to support the complete lifecycle of intellectual property (IP) generated within the university.

The platform shall facilitate invention disclosure, ownership determination, patent management, commercialization, startup support, industry engagement, and revenue distribution.

The system shall consist of:

1. Public-Facing TTO Website
2. Internal Intellectual Property Management Portal
3. Workflow Management System
4. Administrative Dashboard & Reporting System

The objective is to ensure that the TTO can independently manage all activities without requiring continuous vendor support.

02 SYSTEM ARCHITECTURE

The proposed solution shall consist of two integrated components:

A. Public-Facing Website

The public website shall serve as the primary information and outreach platform.



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Website Menu Structure

Home

Overview of TTO activities, featured technologies, latest updates, announcements, and innovation highlights.

About TTO

Mission, vision, objectives, services, organizational structure, and team information.

IP Policy

University Intellectual Property Policy and related governance documents.

SOP & Guidelines

Operational procedures, inventor guidelines, commercialization guidelines, and process flowcharts.

Forms & Templates

Downloadable forms and agreement templates.

Examples:

- Invention Disclosure Form
- IP Assignment Agreement
- NDA/CDA
- Revenue Sharing Agreement
- Waiver/Release Forms
- Sponsored Research Templates

Invention Disclosure

Instructions and access point for online disclosure submission.

Industry Collaboration

Information regarding:

- Sponsored research
- Licensing opportunities
- Consultancy services
- Joint development projects

Startups & Innovation



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Startup showcase, incubation activities, entrepreneurship programs, and innovation success stories.

FAQ

Frequently asked questions relating to patents, ownership, commercialization, and TTO services.

Contact Us

Office information, inquiry forms, appointment requests, and contact details.

B. Internal Intellectual Property Management Portal

The portal shall be accessible only to authorized users and support workflow-driven management of university intellectual property.

Portal Modules

User Registration & Authentication

- User registration
- Login management
- Password recovery
- Role-based permissions

User roles:

- Student
- Faculty
- Staff
- Reviewer
- TTO Officer
- Legal Officer
- Director
- System Administrator

Invention Disclosure Management

- Online disclosure submission



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- Save as draft
 - Supporting document upload
 - Unique disclosure ID generation
 - Status tracking
 - Email notifications
-

Document Management System

- Secure document repository
- Version control
- Search and retrieval
- Access control
- Download history

Supported documents:

- Disclosure forms
 - Assignment agreements
 - NDA/CDA
 - Revenue sharing agreements
 - Sponsored research agreements
 - Licensing agreements
-

IP Assignment Tracking

- Assignment agreement generation
 - Ownership tracking
 - Assignment status monitoring
 - Digital approval workflow
-

Patent Management

- Patent application tracking
- Patent portfolio management
- Filing status monitoring
- Deadline reminders



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- Renewal alerts

Patent statuses:

- Draft
- Filed
- Published
- Examination
- Granted
- Expired
- Abandoned

Commercialization Management

- Technology evaluation
- Licensing opportunities
- Industry engagement tracking
- Startup/spinout management
- Commercialization progress monitoring

Revenue Distribution Management

- Royalty tracking
- Revenue calculations
- Inventor incentive tracking
- Distribution reports

Reports & Analytics

- Disclosure statistics
- Patent statistics
- Commercialization performance
- Startup metrics
- Revenue reports
- Custom dashboards

Export formats:



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- PDF
 - Excel
 - CSV
-

03 INVENTION DISCLOSURE SYSTEM

The system shall provide a structured online disclosure process.

Required fields shall include:

Inventor Information

- Name
- Designation
- Department
- Contact Information

Invention Information

- Title
- Abstract
- Detailed Description
- Technical Field
- Problem Solved
- Novel Features

Commercial Information

- Potential Applications
- Industry Sector
- Existing Alternatives

Research Information

- Funding Source
- Sponsor Information
- Project Reference

Supporting Documents



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- PDF Upload
 - Images
 - Drawings
 - Supporting Files
-

04 WORKFLOW AUTOMATION REQUIREMENTS

The system shall automate the IP management lifecycle.

Invention Disclosure Workflow

Research/Innovation Created



Disclosure Submitted



TTO Review



Ownership Determination



Patentability Evaluation



IP Committee Review



Decision



Patent Filing



Commercialization

Ownership Determination Workflow

Disclosure Received



Review Inventor Status



Review Funding Source



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↓
Review Resource Utilization

↓
Determine Ownership

↓
Execute Agreements

Possible Outcomes:

- University Ownership
- Inventor Ownership
- Joint Ownership
- Sponsored Research Ownership

Patent Filing Workflow

Positive Evaluation

↓
Patent Filing Approval

↓
Patent Drafting

↓
Inventor Review

↓
Patent Application Submission

↓
Patent Prosecution

↓
Patent Grant / Closure

Commercialization Workflow

Protected IP

↓
Market Assessment

↓
Industry Engagement



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↓
Licensing / Startup Decision
↓
Agreement Execution
↓
Revenue Generation

Revenue Distribution Workflow

Revenue Received
↓
Verify Revenue Source
↓
Calculate Distribution Shares
↓
Approval
↓
Distribution to Beneficiaries
↓
Record Maintenance

05 TECHNOLOGY PORTFOLIO & LICENSING

The public website shall maintain a searchable technology portfolio.

Each technology listing shall include:

- Technology Title
- Description
- Industry Sector
- Development Stage
- Benefits
- Licensing Availability
- Contact Information

Industry users shall be able to submit expressions of interest directly through the website.



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06 STARTUP & INNOVATION MANAGEMENT

The platform shall maintain records of university-affiliated startups.

Fields:

- Startup Name
- Founders
- Faculty Advisor
- Technology Used
- Incorporation Date
- Funding Status
- Website

The system shall also provide a public startup showcase.

07 INDUSTRY COLLABORATION PORTAL

External organizations shall be able to submit:

- Sponsored research requests
- Licensing inquiries
- Consultancy requests
- Collaborative research proposals
- Technology scouting requests

All submissions shall be routed to designated TTO personnel.

08 SECURITY REQUIREMENTS

The system shall implement:

- HTTPS/SSL encryption
- Role-based access control
- Secure file storage



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- Audit logs
- Automated backups
- Activity monitoring

The solution must comply with university cybersecurity policies.

09 USER EXPERIENCE REQUIREMENTS

The platform shall be:

- Mobile responsive
- User-friendly
- Accessible
- Searchable
- Optimized for low-bandwidth environments

The portal shall include dashboard-based navigation for all user roles.

10 ADMIN INDEPENDENCE (CRITICAL REQUIREMENT)

The vendor must ensure complete administrative independence.

The TTO team must be capable of:

- Managing disclosures
- Managing patents
- Managing agreements
- Managing technologies
- Publishing website content
- Generating reports
- Managing users

without requiring vendor intervention.

11 TRAINING REQUIREMENTS



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Vendor shall provide:

- Minimum two training sessions
- Administrator training
- End-user training
- PDF documentation
- Video tutorials

12 DELIVERABLES

1. Fully functional public-facing TTO website
2. Internal Intellectual Property Management Portal
3. Workflow management system
4. Patent management module
5. Agreement repository
6. Commercialization management module
7. Startup management module
8. Reporting dashboard
9. Source code and database
10. User manuals and training materials
11. Minimum six months post-deployment technical support

All features listed in this document shall be treated as mandatory requirements unless otherwise agreed in writing.